Base:

Base is very thin (10^{-6} m) and very lightly doped region when compared to the other two regions.

Collector:

The main function of collector is to collect the majority charge carriers supplied by the emitter through the base. Hence, collector is made physically larger than the other two as it has to dissipate more power. It is modarately doped.

Note Because of the differing size and the amount of doping, the emitter and collector cannot be interchanged.

Transistor Biasing

The application of suitable DC voltages across the transistor terminals is called biasing. The transistor biasing is done differently for different uses. The different modes of transistor biasing are given below.

Forward Active:

In this bias, the emitter-base junction is forward biased and the collector-base junction is reverse biased. The transistor is in the active mode of operation. In this mode, the transistor functions as an amplifier.

Saturation:

Here, the emitter-base junction and collector-base junction are forward biased. The transistor has a very large flow of currents across the junctions. In this mode, transistor is used as a closed switch.

Cut-off:

In this bias, the emitter-base junction and collector-base junction are reverse biased. Transistor in this mode acts an open switch.

Note In a PNP transistor, base and collector will be negative with respect to emitter indicated by the middle letter N whereas base and collector will be positive in an NPN transistor indicated by the middle letter P.

10.4.1 Transistor circuit configurations

There are three types of circuit connections for operating a transistor based on the terminal that is used in common to both input and output circuits.

i) Common-Base (CB) configuration

The base is common to both the input and output circuits. The schematic and circuit symbol are shown in Figure 10.26(a) and 10.26(b). The input current is the emitter current I_E and the output current is the collector current I_C. The input signal is applied between emitter and base while the output is measured between collector and base.

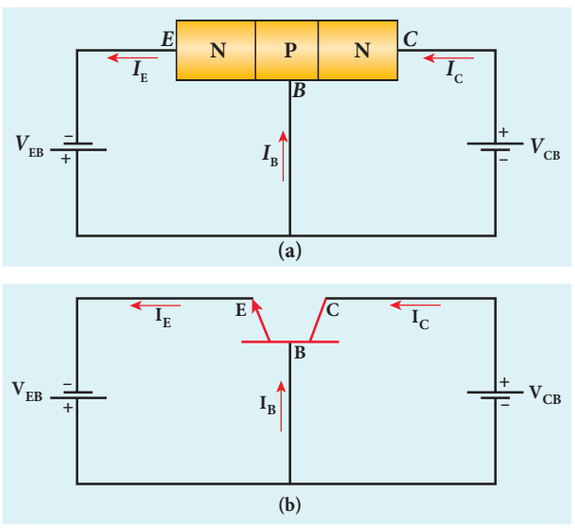


Figure 10.26 NPN transistor in common base configuration (a) Schematic circuit diagram (b) Circuit symbol

XII_Physics_Vol-2_UNIT-10_Part 1_EM.indd 216 19-12-2022 16:39:00